Tab 1

Master Code

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools">

   <application
       android:allowBackup="true"
       android:dataExtractionRules="@xml/data_extraction_rules"
       android:fullBackupContent="@xml/backup_rules"
       android:icon="@mipmap/ic_launcher"
       android:label="@string/app_name"
       android:roundIcon="@mipmap/ic_launcher_round"
       android:supportsRtl="true"
       android:theme="@style/Theme.MyApplication">
       <activity android:name=".SummaryActivity"/>
       <activity android:name=".SecondActivity"/>
       <activity
           android:name=".MainActivity"
           android:exported="true">
           <intent-filter>
               <action android:name="android.intent.action.MAIN" />

               <category android:name="android.intent.category.LAUNCHER" />
           </intent-filter>
       </activity>
   </application>

</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="16dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <EditText
```

```
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <CheckBox
        android:id="@+id/hosteler"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Hosteler"/>

    <CheckBox
        android:id="@+id/mess"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Mess"/>

    <RadioGroup android:id="@+id/genderGroup"
android:layout_width="match_parent"
        android:layout_height="wrap_content">
        <RadioButton android:id="@+id/male" android:text="Male"
android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
        <RadioButton android:id="@+id/female" android:text="Female"
android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
    </RadioGroup>

    <Button
        android:id="@+id/nextBtn"
        android:text="Next"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplication;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```java
import androidx.core.view.WindowInsetsCompat;
import android.widget.EditText;
import android.widget.CheckBox;
import android.widget.RadioGroup;
import android.widget.RadioButton;
import android.widget.Button;
import android.content.Intent;

public class MainActivity extends AppCompatActivity {

   EditText name;
   CheckBox hosteler, mess;
   RadioGroup genderGroup;
   Button nextBtn;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       name = findViewById(R.id.name);
       hosteler = findViewById(R.id.hosteler);
       mess = findViewById(R.id.mess);
       genderGroup = findViewById(R.id.genderGroup);
       nextBtn = findViewById(R.id.nextBtn);

       // Dependency: Hosteler → Mess auto checked
       hosteler.setOnCheckedChangeListener((buttonView, isChecked) -> {
           if (isChecked) mess.setChecked(true);
       });

       nextBtn.setOnClickListener(v -> {

           String userName = name.getText().toString();

           int selectedId = genderGroup.getCheckedRadioButtonId();
           RadioButton selected = findViewById(selectedId);
           String gender = selected.getText().toString();

           Intent i = new Intent(MainActivity.this, SecondActivity.class);
           i.putExtra("name", userName);
           i.putExtra("gender", gender);
           i.putExtra("hosteler", hosteler.isChecked());
           i.putExtra("mess", mess.isChecked());

           startActivity(i);
       });
   }
}
```

activity_second.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="16dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">



   <ToggleButton
       android:id="@+id/toggle"
       android:textOn="Subscribed"
       android:textOff="Not Subscribed"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"/>

   <Button
       android:id="@+id/backBtn"
       android:text="Back"
       android:background="@color/white"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"/>

   <Button
       android:id="@+id/nextBtn2"
       android:text="Next"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"/>

   <Spinner
       android:id="@+id/courseSpinner"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"/>

   <Button
       android:id="@+id/dateBtn"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Select Date"/>

   <TextView
       android:id="@+id/dateText"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="No date selected"/>
```

```
    <Button
        android:id="@+id/popupBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Popup"/>


</LinearLayout>
```

SecondActivity.java

```
package com.example.myapplication;

import android.app.DatePickerDialog;
import android.content.Intent;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.PopupMenu;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;

import java.util.Calendar;

public class SecondActivity extends AppCompatActivity {

    Spinner spinner;
    ToggleButton toggle;
    Button backBtn, nextBtn, dateBtn, popupBtn;
    TextView dateText;

    String selectedDate = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        // Initialize views
        spinner = findViewById(R.id.courseSpinner);
        toggle = findViewById(R.id.toggle);
        backBtn = findViewById(R.id.backBtn);
```

```
        nextBtn = findViewById(R.id.nextBtn2);
        dateBtn = findViewById(R.id.dateBtn);
        popupBtn = findViewById(R.id.popupBtn);
        dateText = findViewById(R.id.dateText);

        // Spinner setup
        String[] courses = {"CSE", "CCE", "IT"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
                android.R.layout.simple_spinner_item, courses);

adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item
);
        spinner.setAdapter(adapter);

        // DatePicker with constraint
        dateBtn.setOnClickListener(v -> {
            Calendar c = Calendar.getInstance();

            DatePickerDialog dp = new DatePickerDialog(this,
                    (view, y, m, d) -> {
                        selectedDate = d + "/" + (m + 1) + "/" + y;
                        dateText.setText(selectedDate);
                    },
                    c.get(Calendar.YEAR),
                    c.get(Calendar.MONTH),
                    c.get(Calendar.DAY_OF_MONTH));

            dp.getDatePicker().setMinDate(System.currentTimeMillis());

dp.getDatePicker().setMaxDate(System.currentTimeMillis()+1000*60*60*24*7);
            dp.show();
        });

        // Popup Menu
        popupBtn.setOnClickListener(v -> {
            PopupMenu popup = new PopupMenu(this, v);
            popup.getMenuInflater().inflate(R.menu.popup_menu,
popup.getMenu());

            popup.setOnMenuItemClickListener(item -> {
                if (item.getItemId() == R.id.option1) {
                    Toast.makeText(this, "Option 1",
Toast.LENGTH_SHORT).show();
                    return true;
                }
                return false;
            });

            popup.show();
```

```
        });

        // Back button
        backBtn.setOnClickListener(v -> finish());

        // Next button
        nextBtn.setOnClickListener(v -> {

            Intent prev = getIntent();

            String name = prev.getStringExtra("name");
            String gender = prev.getStringExtra("gender");
            boolean hosteler = prev.getBooleanExtra("hosteler", false);
            boolean mess = prev.getBooleanExtra("mess", false);

            String course = spinner.getSelectedItem().toString();
            boolean subscribed = toggle.isChecked();

            Intent i = new Intent(SecondActivity.this, SummaryActivity.class);

            i.putExtra("name", name);
            i.putExtra("gender", gender);
            i.putExtra("hosteler", hosteler);
            i.putExtra("mess", mess);
            i.putExtra("course", course);
            i.putExtra("subscribed", subscribed);
            i.putExtra("date", selectedDate);

            startActivity(i);
        });
    }
}
```

activity_summary.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:padding="16dp"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/resultText"
        android:textSize="18sp"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"/>

    <Button
        android:text="Back"
        android:onClick="goBack"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

SummaryActivity.java

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:padding="16dp"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/resultText"
        android:textSize="18sp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:text="Back"
        android:onClick="goBack"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

keepinmind

README — THINGS TO NOT BREAK WHILE COPY-PASTING

PACKAGE NAME

- First line of every Java file: `package com.example.YOURAPPNAME;`
- Must match what Android Studio generated when you created the project
- If you copy-paste a whole Java file, change this line immediately

---

ID CONSISTENCY (most common mistake)

- Every `android:id="@+id/XYZ"` in XML must match `findViewById(R.id.XYZ)` in Java
- If XML says `@+id/datePicker`, Java must say `findViewById(R.id.datePicker)`
- Spelling and camelCase must match exactly — `datePickerr` ≠ `datePicker`

ACTIVITY NAMES

- Class name in Java file = `public class MainActivity extends AppCompatActivity`
- File name = `MainActivity.java`
- XML layout name (in setContentView) = `R.layout.activity_main`
- XML file = `activity_main.xml` in `res/layout/`
- All four must align. Android Studio auto-creates these if you use New → Activity.

MANIFEST: REGISTER EVERY ACTIVITY

- Any new activity you create = add `<activity android:name=".YourActivityName" />` to AndroidManifest.xml
- Only the LAUNCHER activity gets the `<intent-filter>` block
- If you forget this → app crashes on navigation

TOOLBAR vs DEFAULT ACTION BAR

- If you use a `<Toolbar>` in XML + `setSupportActionBar(toolbar)` in Java: → The theme in AndroidManifest must be `Theme.AppCompat.Light.NoActionBar` → Without this, you get two toolbars (the default + yours) and it looks broken
- If you don't want a toolbar, just remove it from XML and don't call setSupportActionBar

---

MENU FILE LOCATION

- Menu XML goes in `res/menu/` (not `res/layout/`)
- You may need to create the `menu` folder: right-click `res` → New → Android Resource Directory → type `menu`
- The menu id in XML (e.g. `R.menu.display_menu`) must match the filename (`display_menu.xml`)

INTENT EXTRAS: KEY MUST MATCH EXACTLY

- Sending: `intent.putExtra("date", dateStr)`
- Receiving: `intent.getStringExtra("date")`
- The string key `"date"` must be identical on both sides
- Typo in the key = null value on receiving end (no crash, just blank data)

CALENDAR MONTHS ARE 0-INDEXED

- January = 0, February = 1, ... December = 11

- When displaying: always use `(month + 1)`
- When comparing with Calendar.getInstance(): also 0-indexed, so they're consistent

---

DATEPICKER setMinDate / setMaxDate

- These use milliseconds (Unix timestamp)
- `System.currentTimeMillis() - 1000` = today minus 1 second (safe "today or after" minimum)
- Always subtract 1000ms otherwise sometimes today's date is not selectable
- Call setMinDate AFTER `datePicker = findViewById(...)`, not before

CHECKBOX LISTENERS FIRING EACH OTHER (infinite loop)

- If checkboxCRA's listener unchecks checkboxOE, and checkboxOE's listener unchecks checkboxCRA...
- This can cause a loop. Android usually handles it gracefully but to be safe:
  checkboxCRA.setOnCheckedChangeListener((btn, isChecked) -> { if (isChecked) { checkboxOE.setOnCheckedChangeListener(null); // detach checkboxOE.setChecked(false); // re-attach after setupCheckboxListeners(); }});
-
- Simpler safe approach: just set checked state and don't worry (Android usually won't loop)

TOGGLE BUTTON + CHECKBOX DEPENDENCY

- Set the toggleButton listener BEFORE you set checkbox listeners
- Order matters: if checkboxMess listener fires before toggle listener is set, behavior is undefined

---

NULL POINTER — COMMON CAUSES

1. `findViewById` called before `setContentView` → ALWAYS call setContentView first
2. Wrong ID → returns null → crash on `.setOnClickListener()`
3. Using a view from a different layout file than the one set in setContentView

---

CHANGING COLORS — COMMON MISTAKES

- `setBackgroundColor()` takes an int (the color value), NOT a string
    - WRONG: `btn.setBackgroundColor("#FF0000")` → compile error
    - RIGHT: `btn.setBackgroundColor(Color.parseColor("#FF0000"))`
    - ALSO RIGHT: `btn.setBackgroundColor(Color.RED)`
- For LinearLayout root background: give the layout an `android:id` in XML first, then `findViewById` it in Java like any other view
- `setBackgroundColor()` removes any existing drawable/style background (e.g., rounded corners). If you want to keep styling, use a drawable instead — but for lab purposes, `setBackgroundColor` is fine
- `Color.parseColor()` requires exactly 6 hex digits: `#FF0000` not `#F00`

MENU SETUP — COMMON MISTAKES

- Forgetting to create `res/menu/` folder → file won't be found, compile error
- Using Toolbar menu WITHOUT setting `NoActionBar` theme → two toolbars appear
- Using `onCreateOptionsMenu` + `onOptionsItemSelected` for a PopupMenu → won't work, these methods are only for Toolbar menus
- Forgetting `setSupportActionBar(toolbar)` in Java when using Toolbar → menu never shows

- `showAsAction` in menu XML needs `app:` prefix (not `android:`) when using AppCompat:
    - WRONG: `android:showAsAction="never"`
    - RIGHT: `app:showAsAction="never"` (and `xmlns:app` must be declared in the `<menu>` tag)
- `Alt + Enter` on the red word → Android Studio suggests the import

- Common ones to add manually if needed:

  import android.widget.DatePicker;
- import android.widget.ToggleButton;
- import android.widget.Spinner;
- import android.widget.CheckBox;
- import android.widget.Button;
- import android.widget.TextView;
- import android.widget.Toast;
- import android.widget.ArrayAdapter;
- import android.widget.PopupMenu;
- import android.widget.AdapterView; // for Spinner listener
- import android.content.Intent;
- import android.graphics.Color; // for color changes
- import android.view.Menu;
- import android.view.MenuItem;
- import androidx.appcompat.app.AlertDialog;
- import androidx.appcompat.widget.Toolbar;
- import androidx.core.content.ContextCompat; // for R.color.xxx
- import java.util.Calendar;
-

QUICK SANITY CHECKLIST BEFORE RUNNING

- [ ] Package name is correct in all Java files
- [ ] All activities registered in AndroidManifest.xml
- [ ] Theme is NoActionBar (if using Toolbar menu)
- [ ] All XML ids match Java findViewById calls
- [ ] Menu file is in `res/menu/` not `res/layout/`
- [ ] Menu XML uses `app:showAsAction` not `android:showAsAction`
- [ ] Intent extra keys match on send and receive side
- [ ] `(month + 1)` when displaying month from DatePicker

- [ ] setMinDate called with `System.currentTimeMillis() - 1000`
- [ ] Color.parseColor() used with full 6-digit hex `#RRGGBB`
- [ ] Root layout has an id if you're changing its background color
- [ ] No import errors (red lines) in Java files
-

manifest&menu

MANIFEST & MENU FILES CHEATSHEET

---

1. AndroidManifest.xml

Located at: `app/src/main/AndroidManifest.xml` Every Activity you create MUST be registered here.

Template (2 activities)

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.myapp">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

        <!-- PAGE 1 (launcher / entry point) -->
        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <!-- PAGE 2 -->
        <activity android:name=".DisplayActivity"
            android:exported="false" />

    </application>

</manifest>
```

Key rules:

- `android:exported="true"` → ONLY on the LAUNCHER activity
- `android:exported="false"` → on all other activities
- Theme must be `NoActionBar` if you're using a Toolbar manually
- If you add a 3rd activity, add another `<activity>` block

---

2. Menu XML File

Located at: `res/menu/display_menu.xml` You must CREATE this folder: right-click `res` → New → Android Resource Directory → Resource type: `menu` Then create file: `display_menu.xml`

Template: 2-item menu (Reset + Static Data)

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">

    <item
        android:id="@+id/menu_reset"
        android:title="Reset Form"
        app:showAsAction="never" />

    <item
        android:id="@+id/menu_static_data"
        android:title="Static Data"
        app:showAsAction="never" />

</menu>
```

Modification: Enter Date + Go Back (for Q2 style)

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">

    <item
        android:id="@+id/menu_enter_date"
        android:title="Enter Date"
        app:showAsAction="never" />
```

```
    <item
        android:id="@+id/menu_go_back"
        android:title="Go Back and Edit"
        app:showAsAction="never" />

</menu>
```

> `showAsAction="never"` means items appear in the overflow (3-dot) menu. Use `showAsAction="always"` to show as icon in toolbar.

---

3. colors.xml — Define Colors for the App

> Located at: `res/values/colors.xml` Android Studio creates this automatically. You can add your own named colors here.

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="colorPrimary">#6200EE</color>
    <color name="colorPrimaryDark">#3700B3</color>
    <color name="colorAccent">#03DAC5</color>

    <!-- Custom colors — add whatever you need -->
    <color name="dangerRed">#F44336</color>
    <color name="safeGreen">#4CAF50</color>
    <color name="warningYellow">#FFEB3B</color>
    <color name="lightRedBg">#FFCDD2</color>
    <color name="lightGreenBg">#C8E6C9</color>
</resources>
```

> To use in Java: `ContextCompat.getColor(this, R.color.dangerRed)` To use in XML: `android:background="@color/dangerRed"`

Using ContextCompat in Java (safer than Color.parseColor)

```
import androidx.core.content.ContextCompat;

// Instead of Color.parseColor("#F44336"), you can do:
btnSubmit.setBackgroundColor(ContextCompat.getColor(this, R.color.dangerRed));
```

---

4. strings.xml (optional but useful)

Located at: `res/values/strings.xml`

```
<resources>
   <string name="app_name">My App</string>
</resources>
```

---

5. HOW TO CREATE MENU FILES — Step by Step in Android Studio

1. In the Project panel (left side), expand: app → res
2. Right-click on res
3. Click: New → Android Resource Directory
4. In the dialog:
   Directory name: menu
   Resource type: menu ← select from dropdown
   Click OK
5. Now right-click on the new res/menu folder
6. Click: New → Menu Resource File
7. Name it: display_menu (no spaces, no .xml — Studio adds .xml)
8. Click OK
9. The file opens — paste your menu XML inside the <menu> tag

If `res/menu` already exists, skip to step 5.

---

6. Checklist: What to register/create for each feature

Feature	File(s) to touch
New Activity	AndroidManifest.xml + new Java file + new XML layout
Toolbar menu	res/menu/your_menu.xml + Java onCreateOptionsMenu

Popup menu (button)	res/menu/your_menu.xml + Java PopupMenu
Custom colors	res/values/colors.xml
New string resource	res/values/strings.xml (optional)

XMLComponents

XML COMPONENTS CHEATSHEET

> All XML goes in `res/layout/activity_main.xml` (Page 1) or `res/layout/activity_display.xml` (Page 2) Root layout is usually `ConstraintLayout` or `LinearLayout (vertical)`

BOILERPLATE: LinearLayout Root (use this as outer wrapper)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- PUT COMPONENTS HERE -->

</LinearLayout>
```

---

1. DATE PICKER (Inline, always visible)

```
<DatePicker
    android:id="@+id/datePicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:calendarViewShown="true"
    android:spinnersShown="false" />
```

Modification: spinner style (compact)

```
<DatePicker
    android:id="@+id/datePicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:calendarViewShown="false"
```

```
    android:spinnersShown="true" />
```

---

2. TOGGLE BUTTON

```
<ToggleButton
    android:id="@+id/toggleButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textOn="Hosteller"
    android:textOff="Off-Campus"
    android:checked="false" />
```

Modification: Emergency ON/OFF

```
<ToggleButton
    android:id="@+id/toggleButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textOn="Emergency ON"
    android:textOff="Emergency OFF"
    android:checked="false" />
```

---

3. SPINNER (Dropdown)

```
<Spinner
    android:id="@+id/spinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />
```

Items are set in Java, not XML (see Java section)

---

4. CHECKBOX

```
<CheckBox
    android:id="@+id/checkboxLab"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```
    android:text="Lab" />

<CheckBox
    android:id="@+id/checkboxCRA"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="CRA" />

<CheckBox
    android:id="@+id/checkboxOE"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="OE" />

<CheckBox
    android:id="@+id/checkboxMess"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Mess" />
```

---

5. BUTTON (Submit / Clear / Back)

```
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit" />

<Button
    android:id="@+id/btnClear"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Clear" />

<Button
    android:id="@+id/btnBack"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Back" />
```

---

6. TEXTVIEW (Display labels / output)

```
<TextView
    android:id="@+id/tvResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Result will appear here"
    android:textSize="16sp" />
```

---

7. TOOLBAR (in activity_display.xml, above everything)

```
<androidx.appcompat.widget.Toolbar
    android:id="@+id/toolbar"
    android:layout_width="match_parent"
    android:layout_height="?attr/actionBarSize"
    android:background="?attr/colorPrimary"
    android:title="Details" />
```

Also needs menu XML file — see MANIFEST_AND_MENU.md

---

8. COMPLETE PAGE 1 EXAMPLE (Fire Safety App style)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Date:" />

    <DatePicker
        android:id="@+id/datePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:calendarViewShown="true"
        android:spinnersShown="false" />
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Alert Mode:" />

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Emergency ON"
        android:textOff="Emergency OFF"
        android:checked="false" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Zone:" />

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Submit" />

</LinearLayout>
```

---

9. COMPLETE PAGE 2 EXAMPLE

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <androidx.appcompat.widget.Toolbar
```

```
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        android:background="?attr/colorPrimary"
        android:title="Details" />

    <TextView
        android:id="@+id/tvDate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Date: " />

    <TextView
        android:id="@+id/tvZone"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Zone: " />

    <TextView
        android:id="@+id/tvToggle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Alert: " />

    <Button
        android:id="@+id/btnBack"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Back" />

</LinearLayout>
```

JavaComponents

JAVA COMPONENTS CHEATSHEET

MainActivity.java = Page 1 | DisplayActivity.java = Page 2 Always extend AppCompatActivity

---

BOILERPLATE: Activity Shell

```
package com.example.myapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // YOUR CODE HERE
    }
}
```

---

1. DATE PICKER — Basic Setup + Get Value

```
// Declare
DatePicker datePicker;

// In onCreate
datePicker = findViewById(R.id.datePicker);

// Get selected date values
int day   = datePicker.getDayOfMonth();
int month = datePicker.getMonth();       // 0-indexed! January = 0
int year  = datePicker.getYear();

String dateStr = day + "/" + (month + 1) + "/" + year;
```

Modification: Set min date = today

```
// In onCreate, after finding the view
datePicker.setMinDate(System.currentTimeMillis() - 1000);
```

Modification: Set max date = today + 20 days

```
Calendar maxCal = Calendar.getInstance();
maxCal.add(Calendar.DAY_OF_MONTH, 20);
datePicker.setMaxDate(maxCal.getTimeInMillis());
```

Modification: Date >= today validation (manual check in submit)

```
Calendar selected = Calendar.getInstance();
selected.set(datePicker.getYear(), datePicker.getMonth(), datePicker.getDayOfMonth());
selected.set(Calendar.HOUR_OF_DAY, 0);
selected.set(Calendar.MINUTE, 0);
selected.set(Calendar.SECOND, 0);
selected.set(Calendar.MILLISECOND, 0);

Calendar today = Calendar.getInstance();
today.set(Calendar.HOUR_OF_DAY, 0);
today.set(Calendar.MINUTE, 0);
today.set(Calendar.SECOND, 0);
today.set(Calendar.MILLISECOND, 0);

if (selected.before(today)) {
    Toast.makeText(this, "Date must be today or later", Toast.LENGTH_SHORT).show();
    return;
}
```

Required imports for Calendar

```
import java.util.Calendar;
```

---

2. TOGGLE BUTTON — Setup + Read State

```
// Declare
ToggleButton toggleButton;

// In onCreate
```

```
toggleButton = findViewById(R.id.toggleButton);

// Read state
boolean isOn = toggleButton.isChecked();
String state = isOn ? "Emergency ON" : "Emergency OFF";
// OR for residence:
String residenceType = toggleButton.isChecked() ? "Hosteller" : "Off-Campus";
```

Modification: Listener (react immediately when toggled)

```
toggleButton.setOnCheckedChangeListener((buttonView, isChecked) -> {
    if (isChecked) {
        // e.g., auto-check Mess checkbox
        checkboxMess.setChecked(true);
    } else {
        checkboxMess.setChecked(false);
    }
});
```

---

3. SPINNER — Setup with Items

```
// Declare
Spinner spinner;

// In onCreate
spinner = findViewById(R.id.spinner);

String[] zones = {"Zone A", "Zone B", "Zone C"};
ArrayAdapter<String> adapter = new ArrayAdapter<>(
    this,
    android.R.layout.simple_spinner_item,
    zones
);
adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
spinner.setAdapter(adapter);

// Read selected value
String selectedZone = spinner.getSelectedItem().toString();
```

Required imports

```
import android.widget.ArrayAdapter;
import android.widget.Spinner;
```

---

4. CHECKBOX — Setup + Read State

```
// Declare
CheckBox checkboxLab, checkboxCRA, checkboxOE, checkboxMess;

// In onCreate
checkboxLab  = findViewById(R.id.checkboxLab);
checkboxCRA  = findViewById(R.id.checkboxCRA);
checkboxOE   = findViewById(R.id.checkboxOE);
checkboxMess = findViewById(R.id.checkboxMess);

// Read state
boolean labChecked  = checkboxLab.isChecked();
boolean craChecked  = checkboxCRA.isChecked();

// Build selected list string
StringBuilder sb = new StringBuilder();
if (checkboxLab.isChecked())  sb.append("Lab ");
if (checkboxCRA.isChecked())  sb.append("CRA ");
if (checkboxOE.isChecked())   sb.append("OE ");
if (checkboxMess.isChecked()) sb.append("Mess ");
String selected = sb.toString().trim();
```

Modification: CRA vs OE mutual exclusion

```
checkboxOE.setOnCheckedChangeListener((buttonView, isChecked) -> {
    if (isChecked) checkboxCRA.setChecked(false);
});

checkboxCRA.setOnCheckedChangeListener((buttonView, isChecked) -> {
    if (isChecked) checkboxOE.setChecked(false);
});
```

Modification: Hosteller → auto-check Mess + prevent uncheck

```
toggleButton.setOnCheckedChangeListener((buttonView, isChecked) -> {
    if (isChecked) {
        checkboxMess.setChecked(true);
```

```
        checkboxMess.setEnabled(false); // prevent unchecking
    } else {
        checkboxMess.setEnabled(true);
        checkboxMess.setChecked(false);
    }
});
```

---

5. ALERT DIALOG (Confirmation popup)

```
new AlertDialog.Builder(this)
    .setTitle("Confirm")
    .setMessage("Are you sure you want to submit?")
    .setPositiveButton("Yes", (dialog, which) -> {
        // Confirmed — do the thing (e.g., navigate)
        Intent intent = new Intent(MainActivity.this, DisplayActivity.class);
        intent.putExtra("date", dateStr);
        intent.putExtra("zone", selectedZone);
        intent.putExtra("toggle", state);
        startActivity(intent);
    })
    .setNegativeButton("No", (dialog, which) -> {
        dialog.dismiss(); // stay on same page
    })
    .show();
```

Required imports

```
import androidx.appcompat.app.AlertDialog;
```

---

6. INTENT — Navigate + Pass Data

Send data (Page 1)

```
Intent intent = new Intent(MainActivity.this, DisplayActivity.class);
intent.putExtra("date", dateStr);
intent.putExtra("zone", selectedZone);
intent.putExtra("toggleState", state);
intent.putExtra("checkboxes", selected); // String with checked items
startActivity(intent);
```

Receive data (Page 2 — in onCreate)

```
Intent intent = getIntent();
String date        = intent.getStringExtra("date");
String zone        = intent.getStringExtra("zone");
String toggleState = intent.getStringExtra("toggleState");
String checkboxes  = intent.getStringExtra("checkboxes");

tvDate.setText("Date: " + date);
tvZone.setText("Zone: " + zone);
tvToggle.setText("State: " + toggleState);
```

---

7. BACK BUTTON

```
btnBack = findViewById(R.id.btnBack);
btnBack.setOnClickListener(v -> finish()); // finish() returns to previous activity
```

---

8. CLEAR BUTTON

```
btnClear.setOnClickListener(v -> {
    toggleButton.setChecked(false);
    checkboxLab.setChecked(false);
    checkboxCRA.setChecked(false);
    checkboxOE.setChecked(false);
    checkboxMess.setChecked(false);
    spinner.setSelection(0);
    // DatePicker resets to today automatically if you set minDate again
    datePicker.setMinDate(System.currentTimeMillis() - 1000);
});
```

---

9. TOOLBAR + OPTIONS MENU (Page 2)

In DisplayActivity.java — setSupportActionBar

```
// In onCreate
Toolbar toolbar = findViewById(R.id.toolbar);
```

```
setSupportActionBar(toolbar);
```

Inflate the menu

```
@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.display_menu, menu);
    return true;
}
```

Handle menu item clicks

```
@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.menu_reset) {
        Intent intent = new Intent(DisplayActivity.this, MainActivity.class);
        intent.addFlags(Intent.FLAG_ACTIVITY_CLEAR_TOP); // clears back stack
        startActivity(intent);
        return true;

    } else if (id == R.id.menu_static_data) {
        // Show static info in a dialog
        new AlertDialog.Builder(this)
            .setTitle("Zone Info")
            .setMessage("Zone A: North\nZone B: South\nZone C: East")
            .setPositiveButton("OK", null)
            .show();
        return true;
    }

    return super.onOptionsItemSelected(item);
}
```

Required imports for Menu

```
import android.view.Menu;
import android.view.MenuItem;
import androidx.appcompat.widget.Toolbar;
```

---

10. POPUP MENU (Button-triggered, alternative to toolbar menu)

```
Button btnMenu = findViewById(R.id.btnMenu);
btnMenu.setOnClickListener(v -> {
    PopupMenu popupMenu = new PopupMenu(DisplayActivity.this, v);
    popupMenu.getMenuInflater().inflate(R.menu.display_menu, popupMenu.getMenu());
    popupMenu.setOnMenuItemClickListener(item -> {
        int id = item.getItemId();
        if (id == R.id.menu_enter_date) {
            tvDate.setText("Date: " + receivedDate);
            return true;
        } else if (id == R.id.menu_go_back) {
            finish();
            return true;
        }
        return false;
    });
    popupMenu.show();
});
```

Required imports

```
import android.widget.PopupMenu;
```

---

11. TOAST (Quick message)

```
Toast.makeText(this, "Your message here", Toast.LENGTH_SHORT).show();
// LENGTH_SHORT or LENGTH_LONG
```

---

12. CHANGING COLORS DYNAMICALLY

> All color changes happen in Java. You reference views and call `.setBackgroundColor()`, `.setTextColor()`, etc. Use `Color.parseColor("#RRGGBB")` for hex colors, or `Color.RED` / `Color.GREEN` / etc. for named colors.

Import needed

```
import android.graphics.Color;
```

12a. Change Button Background Color on Click

```
btnSubmit.setOnClickListener(v -> {
    btnSubmit.setBackgroundColor(Color.parseColor("#FF0000")); // turns red
});
```

Modification: Toggle between two colors on click

```
// Declare a flag at class level
boolean isRed = false;

// In onClick
btnSubmit.setOnClickListener(v -> {
    if (isRed) {
        btnSubmit.setBackgroundColor(Color.parseColor("#4CAF50")); // green
    } else {
        btnSubmit.setBackgroundColor(Color.parseColor("#F44336")); // red
    }
    isRed = !isRed;
});
```

12b. Change Layout / Screen Background Color on Button Click

```
// In XML, give the root layout an id:
// android:id="@+id/rootLayout"

// In Java
LinearLayout rootLayout = findViewById(R.id.rootLayout);

btnSubmit.setOnClickListener(v -> {
    rootLayout.setBackgroundColor(Color.parseColor("#FFEB3B")); // yellow background
});
```

12c. Change TextView Text Color

```
tvResult.setTextColor(Color.parseColor("#E91E63")); // pink text
// OR
tvResult.setTextColor(Color.RED);
```

```
tvResult.setTextColor(Color.GREEN);
tvResult.setTextColor(Color.BLUE);
```

12d. Change Color Based on Toggle State

```
toggleButton.setOnCheckedChangeListener((buttonView, isChecked) -> {
    if (isChecked) {
        toggleButton.setBackgroundColor(Color.parseColor("#F44336")); // red = danger ON
        rootLayout.setBackgroundColor(Color.parseColor("#FFCDD2")); // light red bg
    } else {
        toggleButton.setBackgroundColor(Color.parseColor("#4CAF50")); // green = safe
        rootLayout.setBackgroundColor(Color.parseColor("#C8E6C9")); // light green bg
    }
});
```

12e. Change Color Based on Spinner Selection

```
spinner.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
    @Override
    public void onItemSelected(AdapterView<?> parent, android.view.View view, int position,
long id) {
        switch (position) {
            case 0: rootLayout.setBackgroundColor(Color.parseColor("#BBDEFB")); break; // Zone
A = blue
            case 1: rootLayout.setBackgroundColor(Color.parseColor("#C8E6C9")); break; // Zone B
= green
            case 2: rootLayout.setBackgroundColor(Color.parseColor("#FFE0B2")); break; // Zone C
= orange
        }
    }
    @Override
    public void onNothingSelected(AdapterView<?> parent) {}
});
```

Required extra import for Spinner listener

```
import android.widget.AdapterView;
```

12f. Change Color Based on Checkbox State

```
checkboxLab.setOnCheckedChangeListener((btn, isChecked) -> {
    if (isChecked) {
        checkboxLab.setTextColor(Color.parseColor("#4CAF50")); // green when selected
    } else {
        checkboxLab.setTextColor(Color.parseColor("#000000")); // black when unselected
    }
});
```

12g. Reset All Colors (in Clear button)

```
btnClear.setOnClickListener(v -> {
    rootLayout.setBackgroundColor(Color.WHITE);
    btnSubmit.setBackgroundColor(Color.parseColor("#6200EE")); // default purple
    tvResult.setTextColor(Color.BLACK);
    // reset checkboxes, toggles etc...
});
```

COMMON COLOR HEX CODES (quick reference)

Color	Hex Code	Android Named
Red	#F44336	Color.RED
Green	#4CAF50	Color.GREEN
Blue	#2196F3	Color.BLUE
Yellow	#FFEB3B	Color.YELLOW
Orange	#FF9800	—
White	#FFFFFF	Color.WHITE
Black	#000000	Color.BLACK
Light Red	#FFCDD2	—

Light Green	#C8E6C9	—
Light Blue	#BBDEFB	—
Light Yellow	#FFF9C4	—

---

13. MENU — COMPLETE SETUP GUIDE (Step by Step)

There are TWO types of menus. Choose one based on the question.

TYPE 1: TOOLBAR MENU (3-dot overflow in top bar)

When to use: Question says "Options Menu", "Toolbar Menu", "Menu in action bar"

Step 1 — XML: Add Toolbar to your layout

```
<androidx.appcompat.widget.Toolbar
    android:id="@+id/toolbar"
    android:layout_width="match_parent"
    android:layout_height="?attr/actionBarSize"
    android:background="?attr/colorPrimary"
    android:title="My Screen" />
```

Step 2 — Manifest: Set NoActionBar theme (or you get two toolbars)

```
android:theme="@style/Theme.AppCompat.Light.NoActionBar"
```

Step 3 — Create menu folder + file

- Right-click `res` → New → Android Resource Directory → type: `menu` → OK
- Right-click `res/menu` → New → Menu Resource File → name: `display_menu` → OK

Step 4 — Write display_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
```

```
    <item android:id="@+id/menu_option1" android:title="Option 1" app:showAsAction="never" />
    <item android:id="@+id/menu_option2" android:title="Option 2" app:showAsAction="never" />

</menu>
```

Step 5 — Java: Connect toolbar + override two methods

```
// In onCreate:
Toolbar toolbar = findViewById(R.id.toolbar);
setSupportActionBar(toolbar);

// Outside onCreate (new methods in the class):
@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.display_menu, menu);
    return true;
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.menu_option1) {
        // do something
        Toast.makeText(this, "Option 1 clicked", Toast.LENGTH_SHORT).show();
        return true;

    } else if (id == R.id.menu_option2) {
        finish(); // or navigate somewhere
        return true;
    }

    return super.onOptionsItemSelected(item);
}
```

Step 6 — Required imports

```
import android.view.Menu;
import android.view.MenuItem;
import androidx.appcompat.widget.Toolbar;
```

TYPE 2: POPUP MENU (triggered by a Button)

When to use: Question says "Popup Menu", "Button that shows a menu", or there's no Toolbar

Step 1 — XML: Just a normal Button

```
<Button
   android:id="@+id/btnMenu"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Menu" />
```

Step 2 — Create menu folder + file (same as above)

- `res/menu/display_menu.xml` with your items

Step 3 — Java: PopupMenu on button click

```
Button btnMenu = findViewById(R.id.btnMenu);
btnMenu.setOnClickListener(v -> {
   PopupMenu popupMenu = new PopupMenu(this, v); // 'v' = the button (anchor)
   popupMenu.getMenuInflater().inflate(R.menu.display_menu, popupMenu.getMenu());
   popupMenu.setOnMenuItemClickListener(item -> {
      int id = item.getItemId();

      if (id == R.id.menu_option1) {
         Toast.makeText(this, "Option 1 clicked", Toast.LENGTH_SHORT).show();
         return true;

      } else if (id == R.id.menu_option2) {
         finish();
         return true;
      }

      return false;
   });
   popupMenu.show();
});
```

Step 4 — Required import

```
import android.widget.PopupMenu;
```

NO need for NoActionBar theme with PopupMenu. NO need for onCreateOptionsMenu. Just the button + the click listener.

MENU QUICK DECISION TABLE

Situation	Use	Needs NoActionBar?	Needs onCreateOptionsMenu?
Question says "Options Menu"	Toolbar Menu	YES	YES
Question says "Popup Menu"	PopupMenu on Button	NO	NO
Question says "menu button"	PopupMenu on Button	NO	NO
No toolbar in layout	PopupMenu on Button	NO	NO
Toolbar widget in layout	Toolbar Menu	YES	YES

---

14. COMPLETE MainActivity.java EXAMPLE (Fire Safety App)

```
package com.example.firesafety;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    DatePicker datePicker;
```

```
ToggleButton toggleButton;
Spinner spinner;
Button btnSubmit;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    datePicker   = findViewById(R.id.datePicker);
    toggleButton = findViewById(R.id.toggleButton);
    spinner      = findViewById(R.id.spinner);
    btnSubmit    = findViewById(R.id.btnSubmit);

    // Date constraints: today to today+20
    datePicker.setMinDate(System.currentTimeMillis() - 1000);
    Calendar maxCal = Calendar.getInstance();
    maxCal.add(Calendar.DAY_OF_MONTH, 20);
    datePicker.setMaxDate(maxCal.getTimeInMillis());

    // Spinner setup
    String[] zones = {"Zone A", "Zone B", "Zone C"};
    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
            android.R.layout.simple_spinner_item, zones);
    adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
    spinner.setAdapter(adapter);

    // Submit
    btnSubmit.setOnClickListener(v -> {
        int day = datePicker.getDayOfMonth();
        int month = datePicker.getMonth();
        int year = datePicker.getYear();
        String dateStr = day + "/" + (month + 1) + "/" + year;
        String zone = spinner.getSelectedItem().toString();
        String state = toggleButton.isChecked() ? "Emergency ON" : "Emergency OFF";

        new AlertDialog.Builder(this)
            .setTitle("Confirm")
            .setMessage("Date: " + dateStr + "\nZone: " + zone + "\nState: " + state)
            .setPositiveButton("Confirm", (dialog, which) -> {
                Intent intent = new Intent(MainActivity.this, DisplayActivity.class);
                intent.putExtra("date", dateStr);
                intent.putExtra("zone", zone);
                intent.putExtra("toggleState", state);
```

```
                startActivity(intent);
            })
            .setNegativeButton("Cancel", (dialog, which) -> dialog.dismiss())
            .show();
        });
    }
}
```

---

15. COMPLETE DisplayActivity.java EXAMPLE (Fire Safety App)

```
package com.example.firesafety;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import android.content.Intent;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.*;

public class DisplayActivity extends AppCompatActivity {

    TextView tvDate, tvZone, tvToggle;
    Button btnBack;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_display);

        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);

        tvDate   = findViewById(R.id.tvDate);
        tvZone   = findViewById(R.id.tvZone);
        tvToggle = findViewById(R.id.tvToggle);
        btnBack  = findViewById(R.id.btnBack);

        Intent intent = getIntent();
        String date   = intent.getStringExtra("date");
```

```
        String zone   = intent.getStringExtra("zone");
        String state  = intent.getStringExtra("toggleState");

        tvDate.setText("Date: " + date);
        tvZone.setText("Zone: " + zone);
        tvToggle.setText("Alert: " + state);

        btnBack.setOnClickListener(v -> finish());
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.display_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.menu_reset) {
            Intent intent = new Intent(DisplayActivity.this, MainActivity.class);
            intent.addFlags(Intent.FLAG_ACTIVITY_CLEAR_TOP);
            startActivity(intent);
            return true;
        } else if (id == R.id.menu_static_data) {
            new AlertDialog.Builder(this)
                .setTitle("Zone Info")
                .setMessage("Zone A: North City\nZone B: South City\nZone C: East City")
                .setPositiveButton("OK", null)
                .show();
            return true;
        }
        return super.onOptionsItemSelected(item);
    }
}
```

A1 Reference

COMPLETE SOLUTION: Student Residence App (Q2 Style)

> Use this as a full reference when the question has: Toggle + Checkboxes + Constraints + DatePicker + PopupMenu

---

activity_main.xml (Page 1)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Residence Type:" />

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Hosteller"
        android:textOff="Off-Campus"
        android:checked="false" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Facilities:" />

    <CheckBox
        android:id="@+id/checkboxLab"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Lab" />
```

```
<CheckBox
    android:id="@+id/checkboxCRA"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="CRA" />

<CheckBox
    android:id="@+id/checkboxOE"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="OE" />

<CheckBox
    android:id="@+id/checkboxMess"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Mess" />

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Select Date:" />

<DatePicker
    android:id="@+id/datePicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:calendarViewShown="true"
    android:spinnersShown="false" />

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btnClear"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Clear" />

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Submit" />

    </LinearLayout>

</LinearLayout>
```

---

activity_display.xml (Page 2)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/tvResidence"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Residence: " />

    <TextView
        android:id="@+id/tvCheckboxes"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Facilities: " />

    <TextView
        android:id="@+id/tvDate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Date: " />

    <Button
        android:id="@+id/btnMenu"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu" />

</LinearLayout>
```

---

res/menu/display_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menu_enter_date"
        android:title="Enter Date" />

    <item
        android:id="@+id/menu_go_back"
        android:title="Go Back and Edit" />

</menu>
```

MainActivity.java (Page 1)

```
package com.example.studentapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    ToggleButton toggleButton;
    CheckBox checkboxLab, checkboxCRA, checkboxOE, checkboxMess;
    DatePicker datePicker;
    Button btnClear, btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toggleButton  = findViewById(R.id.toggleButton);
        checkboxLab   = findViewById(R.id.checkboxLab);
        checkboxCRA   = findViewById(R.id.checkboxCRA);
```

```
checkboxOE    = findViewById(R.id.checkboxOE);
checkboxMess  = findViewById(R.id.checkboxMess);
datePicker    = findViewById(R.id.datePicker);
btnClear      = findViewById(R.id.btnClear);
btnSubmit     = findViewById(R.id.btnSubmit);

// Date >= today
datePicker.setMinDate(System.currentTimeMillis() - 1000);

// Hosteller → auto-check Mess
toggleButton.setOnCheckedChangeListener((btn, isChecked) -> {
    if (isChecked) {
        checkboxMess.setChecked(true);
        checkboxMess.setEnabled(false);
    } else {
        checkboxMess.setEnabled(true);
        checkboxMess.setChecked(false);
    }
});

// CRA vs OE mutual exclusion
checkboxCRA.setOnCheckedChangeListener((btn, isChecked) -> {
    if (isChecked) checkboxOE.setChecked(false);
});
checkboxOE.setOnCheckedChangeListener((btn, isChecked) -> {
    if (isChecked) checkboxCRA.setChecked(false);
});

// Clear
btnClear.setOnClickListener(v -> {
    toggleButton.setChecked(false);
    checkboxLab.setChecked(false);
    checkboxCRA.setChecked(false);
    checkboxOE.setChecked(false);
    checkboxMess.setChecked(false);
    checkboxMess.setEnabled(true);
    datePicker.setMinDate(System.currentTimeMillis() - 1000);
});

// Submit
btnSubmit.setOnClickListener(v -> {
    String residenceType = toggleButton.isChecked() ? "Hosteller" : "Off-Campus";

    StringBuilder sb = new StringBuilder();
```

```
            if (checkboxLab.isChecked())  sb.append("Lab ");
            if (checkboxCRA.isChecked())  sb.append("CRA ");
            if (checkboxOE.isChecked())   sb.append("OE ");
            if (checkboxMess.isChecked()) sb.append("Mess ");
            String facilities = sb.toString().trim();

            int day   = datePicker.getDayOfMonth();
            int month = datePicker.getMonth();
            int year  = datePicker.getYear();
            String dateStr = day + "/" + (month + 1) + "/" + year;

            Intent intent = new Intent(MainActivity.this, DisplayActivity.class);
            intent.putExtra("residence", residenceType);
            intent.putExtra("facilities", facilities);
            intent.putExtra("date", dateStr);
            startActivity(intent);
        });
    }
}
```

DisplayActivity.java (Page 2)

```
package com.example.studentapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.*;

public class DisplayActivity extends AppCompatActivity {

    TextView tvResidence, tvCheckboxes, tvDate;
    Button btnMenu;
    String receivedDate;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_display);

        tvResidence  = findViewById(R.id.tvResidence);
        tvCheckboxes = findViewById(R.id.tvCheckboxes);
```

```
        tvDate       = findViewById(R.id.tvDate);
        btnMenu      = findViewById(R.id.btnMenu);

        Intent intent = getIntent();
        String residence  = intent.getStringExtra("residence");
        String facilities = intent.getStringExtra("facilities");
        receivedDate      = intent.getStringExtra("date");

        tvResidence.setText("Residence: " + residence);
        tvCheckboxes.setText("Facilities: " + facilities);
        tvDate.setText("Date: " + receivedDate);

        // Popup menu
        btnMenu.setOnClickListener(v -> {
            PopupMenu popupMenu = new PopupMenu(DisplayActivity.this, v);
            popupMenu.getMenuInflater().inflate(R.menu.display_menu, popupMenu.getMenu());
            popupMenu.setOnMenuItemClickListener(item -> {
                int id = item.getItemId();
                if (id == R.id.menu_enter_date) {
                    tvDate.setText("Date: " + receivedDate);
                    return true;
                } else if (id == R.id.menu_go_back) {
                    finish();
                    return true;
                }
                return false;
            });
            popupMenu.show();
        });
    }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.studentapp">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="Student App"
```

```
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light">

        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <activity android:name=".DisplayActivity"
            android:exported="false" />

    </application>

</manifest>
```

NOTE: This example uses default ActionBar theme (not NoActionBar) since there's no Toolbar widget here — only a PopupMenu triggered by a Button. That's why the manifest theme is `Theme.AppCompat.Light` not `NoActionBar`.